PROBLEM 16

Drag on Falling Football

Original statement · Problems.pdf, p. 9

Quadratic Drag

A football has frontal surface area S, mass m, and a drag coefficient C_D. It falls vertically through the air of density ρ. It is assumed that the forces acting on the ball are only gravity and drag. The drag can be modeled as: $D = \frac{1}{2}\rho v^2 S C_D$, with v the speed of the ball. Compute its terminal speed and its position as a function of time.

You may find the following integrals useful:

$$\int \frac{dx}{x_0^2 - x^2} = \frac{1}{x_0}\operatorname{atanh}\left(\frac{x}{x_0}\right); \qquad \int \tanh x\, dx = \ln\cosh x$$

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a football of mass m, frontal area S and drag coefficient C_D falls vertically through air of density ρ. Only gravity and the quadratic drag $D = \frac{1}{2}\rho v^2 S C_D$ act.
- **Constraints:** vertical motion; $\mathrm{CDOF} = 1$, with the coordinate s measured downward from the release point.
- **Initial conditions:** $s(0) = 0, v(0) = 0$. The statement gives $\displaystyle\int \frac{dx}{x_0^2 - x^2} = \frac{1}{x_0}\operatorname{atanh}\frac{x}{x_0}$ and $\int \tanh x\, dx = \ln\cosh x$.

Reference Frames & Vector Bases

- **Inertial frame:** fixed to the air, with Os pointing vertically downward.

Derivation

3.1 Kinematics

$$\mathbf{r} = s\,\mathbf{i}, \qquad \mathbf{v} = \dot{s}\,\mathbf{i} = v\,\mathbf{i}, \qquad \mathbf{a} = \dot{v}\,\mathbf{i}$$

3.2 Forces

- **Weight:** mg downward. **Drag:** $D = \frac{1}{2}\rho v^2 S C_D$, upward because it opposes the downward velocity ($v > 0$). **Inertia forces:** none.

3.3 Equations of Motion

- **Newton's second law:**

$$m\frac{dv}{dt} = mg - \tfrac{1}{2}\rho S C_D v^2$$

One unknown, $v(t)$ (and then $s(t)$ by integration), and one first-order equation.

3.4 Integration and Answers

Terminal velocity

Pedagogical Justification: at the terminal velocity the net force vanishes: $mg = \frac{1}{2}\rho S C_D v_t^2$:

$$\boxed{v_t = \sqrt{\frac{2mg}{\rho S C_D}}}$$

Velocity as a function of time

Pedagogical Justification: using $\frac{1}{2}\rho S C_D = mg/v_t^2$ writes the equation in terms of v_t and allows the given integral:

$$\frac{dv}{dt} = g\left(1 - \frac{v^2}{v_t^2}\right) = \frac{g}{v_t^2}\left(v_t^2 - v^2\right) \implies \frac{dv}{v_t^2 - v^2} = \frac{g}{v_t^2}\,dt$$

- Integrating from $v = 0$ at $t = 0$ (Barrow, with $\operatorname{atanh} 0 = 0$):

$$\left[\frac{1}{v_t}\operatorname{atanh}\frac{v'}{v_t}\right]_0^v = \frac{g}{v_t^2}\int_0^t dt' \implies \operatorname{atanh}\frac{v}{v_t} = \frac{gt}{v_t} \implies \boxed{v(t) = v_t \tanh\left(\frac{gt}{v_t}\right)}$$

The speed increases from 0 (the net force is positive while $v < v_t$) and cannot reach v_t in finite time, because $v \equiv v_t$ is a solution (the terminal state) and solutions are unique; so $|v/v_t| < 1$ and $\operatorname{atanh}(v/v_t)$ is defined for all t; consistently, $\tanh < 1$.

Position as a function of time

Pedagogical Justification: $s(t) = \int_0^t v\,dt'$. With the substitution $u = gt'/v_t$, $du = (g/v_t)\,dt'$, the limits are $u = 0$ at $t' = 0$ and $u = gt/v_t$ at $t' = t$:

$$s(t) = v_t \cdot \frac{v_t}{g}\int_0^{gt/v_t} \tanh u\,du = \frac{v_t^2}{g}\Big[\ln\cosh u\Big]_0^{gt/v_t} = \frac{v_t^2}{g}\left[\ln\cosh\frac{gt}{v_t} - \ln\cosh 0\right]$$

$$\boxed{s(t) = \frac{v_t^2}{g}\ln\left[\cosh\left(\frac{gt}{v_t}\right)\right]} \qquad (\ln\cosh 0 = \ln 1 = 0)$$

Interpretation & Checks

- **Early times ($t \ll v_t/g$):** $\tanh u \approx u$ gives $v \approx gt$; and $\cosh u \approx 1 + u^2/2$, $\ln\cosh u \approx u^2/2$ give $s \approx \frac{1}{2}gt^2$: free fall, the drag is negligible.
- **Late times ($t \gg v_t/g$):** $\tanh u \to 1$, so $v \to v_t$; and $\cosh u \approx \frac{1}{2}e^u$ gives $\ln\cosh u \approx u - \ln 2$, so $s \approx v_t t - \dfrac{v_t^2}{g}\ln 2$: uniform descent with a constant delay.
- **Dimensional check:** $[v_t] = \sqrt{\dfrac{\mathrm{kg\ m\ s^{-2}}}{\mathrm{kg\ m^{-3}m^2}}} = \mathrm{m/s}$; gt/v_t is dimensionless; $[v_t^2/g] = \mathrm{m}$.
- **Check against the official key:** the key of `Problems.pdf` gives the terminal velocity $v_t = \sqrt{2mg/(\rho S C_D)}$, $v(t) = v_t \tanh\left(\frac{g}{v_t}t\right)$ and $s(t) = \frac{v_t^2}{g}\ln\left[\cosh\left(\frac{g}{v_t}t\right)\right]$, identical to the results above. Numerical integration of $\dot v = 1 - v^2$, $\dot s = v$ (units $g = v_t = 1$) up to $t = 1$ gives $v = 0.761594 = \tanh 1$ and $s = 0.433781 = \ln\cosh 1$.

Source: Problems.pdf